

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization, bringing over 4 years of professional experience as a Software Developer and Data Engineer, specializing in building scalable microservices and distributed data pipelines. Possesses a Master of Science in Computer Science with deep expertise in Java, Python, Spark, SQL, and cloud platforms (AWS, Azure). Proven track record in CI/CD automation, large-scale data platforms, and privacy-preserving AI applications, adept at leveraging distributed systems and REST APIs to drive impactful solutions.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Architected and optimized high-performance REST and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, managing over 50,000 daily API requests with 99.99 percent SLA uptime.
- Engineered and automated robust CI/CD pipelines utilizing Jenkins, Maven, and Azure DevOps, facilitating continuous delivery across Azure Cloud and Oracle WebLogic environments.
- Modernized legacy systems by migrating to Log4j 2.x, enhancing monitoring and error handling capabilities which reduced issue resolution time by 20 percent.
- Spearheaded a critical EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams, delivering FHIR-compliant healthcare solutions with effective use of Git, SVN, and JUnit testing.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered robust Java and EJB-based enterprise applications and RESTful web services for key automotive clients.
- Boosted system performance by 90 percent through meticulous SQL query optimization, database tuning, and modernization of legacy Java code.
- Implemented comprehensive CI/CD pipelines using Jenkins and GitLab, integrating automated testing and quality gates for enhanced code integrity.
- Mentored junior developers by conducting thorough code reviews, facilitating debugging sessions, and promoting clean coding practices.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening overall data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving a code coverage of 99 percent.
- Supported release planning and sprint execution using JIRA within Agile development teams, ensuring efficient project delivery.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Terraform) January 2024 - May 2025

- Developed a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real time, supporting rapid data ingestion.
- Built an interactive Flask web application for data exploration featuring dynamic filtering and visualization capabilities.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments and scalable infrastructure.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight, demonstrating infrastructure automation expertise.

Privacy-Preserving AI Email Client (Capstone Project) (T5, LoRA) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform advanced email summarization and classification tasks.
- Applied parameter-efficient fine-tuning using LoRA, supporting low-resource training under strict privacy constraints.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment of AI models.

SKILLS

- Programming Languages: Java, Python, SQL, Go.
- Frameworks & Libraries: Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, JUnit, Flask.
- Web Services: REST APIs, SOAP APIs.
- Cloud Platforms: AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), Google Cloud Platform.
- Databases: PostgreSQL, Oracle Database, IBM DB2, RDS, Elasticsearch, OpenSearch.

- Tools & Technologies: Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, Terraform, GitLab.
- Concepts: ETL Pipelines, Data Warehousing, Data Analytics, Clustering, Boosting, Distributed Systems, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, Privacy-preserving LLM-based applications, Data Ingestion.
- Data Analysis & Visualization: Tableau, QuickSight.